

30	C	initial count rate of source alone, $C_0 = 100 - 20 = 80 \text{ s}^{-1}$ no. of half-lives, $n = \frac{t}{t_{1/2}} = \frac{60}{20} = 3$ $\frac{C}{C_0} = \left(\frac{1}{2}\right)^n$ $C = \left(\frac{1}{2}\right)^n C_0 = \left(\frac{1}{2}\right)^3 (80)$ recorded count rate $= \left(\frac{1}{2}\right)^3 (80) + 20 = 30 \text{ s}^{-1}$